mated irrigation can easily increase yield by 30-40 per cent, provided fertigation schedules are properly maintained. A farmer who has been growing an average of 40-50 metric tonnes yield volume manually at best, can expect production of more than 100 metric tonnes with the help of automation. Additionally, optimised use of water and fertiliser greatly improves the quality of the crop or fruit. With higher yield and better quality of produce, one can ensure much better profits in the market."

Ingole shares the story of their customer Parmeshwar Raut, a landowner based out of Penur village in Solapur district of Maharashtra, who owns 110 acres of cultivation land. The area being large is very difficult to irrigate and maintain manually.

Raut set up a centralised control system provided by Galcon to control the drip irrigation process for their cultivation land. The centralised drip irrigation system operates four pumps throughout the land and performs automated irrigation as per requirement. Since automating irrigation, based on yearly results, the farmer testified an average 40 per cent increase in profit due to increase in crop yield, decrease in fertiliser usage, reduced labour cost and decreased electricity usage.

Save up to 30 per cent of water. Sensor-driven systems ensure that optimal quantity of water is used at the right time, based on the soil moisture content, humidity, rainfall, soil temperature and so on. As a result, automated irrigation can save up to 30 per cent of water usage.

Reduced labour cost, effort and electricity. With automation, agro-owners can save a lot of money, which they otherwise have to invest in manual labour for irrigation. A significant point to focus is erratic power availability. In locations where power is available only during odd hours of the day, hiring a labour for irrigation is not a good idea.

Savings for farmers

Satish K.S., founder & CEO, Flybird, shares how much money a small or medium-scale vegetable farmer with 2-3 crop rotations per year can save, based on the performance of one of their timer-based irrigation controller.

Benefit	Savings
Labour cost reduction	At least ₹ 150 per day. Considering 20 days of labour a month, a farmer can save ₹ 9000 every three months (one crop cycle).
Increased yield volume	Lower crop mortality and higher crop size or quality increases the yield by 15 per cent.
Optimised fertiliser and chemical use	Automated fertigation and chemigation can save 10 per cent fertilisers, 10 per cent weedicide and pesticide.
Lower electricity and water consumption, longer motor life	25-30 per cent of water and electricity can be saved. Pump motors also work better due to precise on and off system by automated controls.
Return on investment	System investments include controller, valves, cabling and installation—based on requirement. Return can be expected within 1-2 crop cycles.

Sensors and automation, on the other hand, resume action as soon as power is back in case the last session of irrigation didn't complete. This ensures optimal irrigation throughout the day. Additionally, manual labour often uses power-driven systems for irrigation, even when these are not required, wasting electricity.

Lesser manual intervention implies fewer errors. Programmed systems distribute the right amount of ingredients at the right time based on environmental readings picked up by sensors using the right amount of power. The process is accurate—facilitating higher quality and quantity.

Bengaluru-based irrigation automation solution provider Flybird Farm Innovations shares the example of its client Gangadhar, who is based in Kolkata and owns a 5-acre farm for grapes cultivation. Gangadhar faced following problems:

1. Power supply availability only for two hours during daytime and night each was impacting his daily routine as he had to be available in the farm, which was 2 kilometres away from his residence, during those odd hours

2. Uneven availability of water caused problems of over-irrigation as well as under-irrigation, which affected the yield volume and quality

The farmer used Flybird's low-cost automation system, which included a two-valve irrigation controller. The system controls valves based on the moisture level and rainfall readings of the soil. It automatically turns on when electricity is available and turns off if power cuts off in the middle of the irrigation. It resumes function whenever the power resumes and completes the pending irrigation. Water volume is also accurately distributed as per the program.

Spend less on fertilisers. Sensors detect the amount of fertiliser and chemicals in the soil. When necessary, the system automatically starts the fertigation process and continues until the required amount of fertiliser has been introduced. This prevents overuse of the product and reduces fertiliser usage by 10-30 per cent.

The final word

An automated system is a big boon both to the agri-community and the national economy. Training and guidance for farm-owners and labourers are often arranged by the manufacturers, so practical know-how is not a challenge. EFY

—Paromik Chakraborty, technical journalist, EFY